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DevOps Engineer

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Technical Skills

* DevOps Engineering
* Containerization
* Container Orchestration
* CI/CD Engineering
* Full Stack Development
* Linux Ecosystem
* Relational Databases
* Web Servers

Work Experience

01.2023 - current

Freelancing

Providing support as DevOps/Platform engineer in various projects.

Kubernetes **Ansible** **Helm Charts** **GitLab CI** **Prometheus Stack** **Loki Stack** **Grafana Alerts** **Renovate** **Infrastructure-as-a-code**

05.2021 - 12.2023

Kubernetes Engineer

Next Kraftwerke GmbH, Cologne, Germany

Project: Setting up secure and reliable multi-tenant kubernetes clusters on-premises in highly regulated and secure environments.

Responsibilities:

- Development of the platform as IaC using terraform and ansible
- Setting up and maintaining CI pipelines within Gitlab and Jenkins
- Setting up and maintaining CD with ArgoCD
- Setting up monitoring, alerting and logging with Prometheus and Loki stacks
- Policy enforcement with Kyverno
- Implementing K8s network policies for each application
- Implementing tests
- Setting up Renovate for automated dependency updates
- Collaborating with developers to better understand their needs
- Preparing learning sessions
- Hosting daily meetings regularly

kubespray **rancher** **MetalLB** **RKE** **Terraform** **Ansible** **GitLab CI** **Argo CD** **Nexus**

10.2017 - 12.2020

Lead Software Developer

GESIS – Leibniz Institute for the Social Sciences, Cologne, Germany

Project: GESIS Notebooks is a free and hosted service for running Jupyter Notebooks requiring no installation, which enables users to get started quickly with their research ideas, prototyping their data science projects, or learning to program languages such as R, Python or Julia.

Responsibilities:

- Planning of the complete software development process
- Setup and maintenance of the servers and the kubernetes cluster

- Deployment and maintenance of a Persistent BinderHub and a public BinderHub (deployment repo: gesiscss/orc)
- Automation of all aspects including image building, chart and configuration updates and deployments
- Initial implementation of the "GESIS Login" system (based on Keycloak)
- Cooperation with open source projects (Jupyter projects) and contributing
- Jupytercon 2020 Talk: "A Persistent BinderHub: Democratizing Access to Computational Resources in the Social Sciences"
- Tutoring in "Introduction to Python" workshop
- Publication of "iLCM - A Virtual Research Infrastructure for Large-Scale Qualitative Data."

Reference: Dr. Arnim Bleier

python | Docker | Kubernetes | Helm | GitLab CI/CD | Prometheus | Grafana | Elastic Stack | Node.js | npm | nginx

07.2016 - 09.2017

Lead Software Developer

GESIS – Leibniz Institute for the Social Sciences, Cologne, Germany

Project: WikiWho is an algorithm to parse the complete set of all historical revisions of a revisioned Wiki document in order to find out who wrote and/or removed and/or reinserted which exact text at token (~word) level at what revision. This means that for every token, its individual add/removal/reintroduction history becomes available.

Responsibilities:

- Speed and accuracy improvement of the existing core WikiWho algorithm
- Realisation of the website, the REST API and a web-based API interface
- Parsing Wikipedia archives for various languages and applying WikiWho on them to generate base data for APIs
- Realisation of an event listener for Wikipedia events, connected to a task queue
- Update and improvement of the WhoColor server and user tools
- Data analyses of authorship, persistence, conflict and editor interactions
- Generation of a data set for the entire English Wikipedia: "TokTrack: A Complete Token Provenance and Change Tracking Dataset for the English Wikipedia"

Reference: Dr. Fabian Flöck

python | Django | Gunicorn | nginx | REST API | Swagger | PostgreSQL | Celery | RabbitMQ | WikiPedia

07.2015 - 06.2016

Software Developer

MIR MEDIA, Cologne, Germany

Backend development of websites for customers such as Elbphilarmonie and Lucerne Festival

python | Django

11.2014 - 03.2015

Software Developer

Induux GmbH, Stuttgart, Germany

Backend development

python | Django

11.2013 - 05.2014

Student Employee

Elektrobit Automotive GmbH, Böblingen, Germany

python | Django | Selenium | REST API

04.2013 - 09.2013

Intern

Elektrobit Automotive GmbH, Böblingen, Germany

python | software versioning | Unit testing | GUI development | CAN Bus | CANoe

11.2012 - 03.2013

Student Employee

PTV AG, Stuttgart, Germany

C++ | Visualization | The Visualization ToolKit

Most Recent Projects

Binder Launches

Binder launches repo contains

- a script to parse binder launch events and save into TimescaleDB
- a frontend to visualize launches using React with Material-UI which is backed by Express.js

React | Material-UI | Express.js | Node.js | npm | python | TimescaleDB | Docker | Helm chart | GitHub Actions

Higher Education

University of Stuttgart, Germany

10.2011 - 05.2014

M.Sc. in Information Technology

Master Thesis at Industrial Automation Systems Institute

Topic: Realization of incremental learning for automated identification using the example of products recognition

- Incremental Learning, Machine Learning
- Classification with Support Vector Machines (SVM)
- Learn++ and Learn++.NC based on SVM
- Python, scikit-learn ML library

Istanbul Technical University, Turkey

09.2006 - 06.2011

B.Sc. in Electronics Engineering (graduation in honor list)

Bachelor Thesis at Embedded System Design Laboratory

Topic: Design and implementation of a server on a graphic processor for secure RFID systems

- Parallel programming with CUDA
- RFID protocol servers

Extracurricular Activities

- 09.2007 - 06.2011 : Member of the ITU Control and Automation Student Society
- 09.2008 - 06.2009 : Committee Member of the ITU Robotic Olympics
- 09.2008 - 01.2009 : Member of the ITU Mountaineering Club

ESIGELEC, Rouen, France

09.2009 - 02.2010

Exchange Semester

Languages

* Turkish (native)
* English (fluent)
* German (B2)

Interests

* Hiking
* Swimming
* Jogging
* Pilates
* Reading
* Meditation
* Wood work
* Bicycling